

第三章: 边值问题的分离变量法和特殊函数(II)

考: 幂级数法解二阶ODE

柱域上的分离变量法: Bessel函数的引出

$$\text{圆盘 } \bar{\Sigma}_a = \{X = (x, y) \in \mathbb{R}^2 : \|X\| < a\}.$$

边缘温度为零, 内部不产生热量, 初始温度 $\phi(X)$, 温度分布 $u(X, t)$ 满足:

$$\begin{cases} \partial_t u = \Delta u & , X \in \bar{\Sigma}_a, t > 0 \\ u(X, 0) = \phi(X), X \in \bar{\Sigma}_a \\ u(X, t) = 0 & , X \in S_a := \partial \bar{\Sigma}_a, t > 0. \end{cases}$$

分离变量法: $u(x, y, t) = \sum_{n,m} V_{nm}(x, y) T_{nm}(t)$

$$\text{其中 } V \text{ 满足 } \begin{cases} \Delta V = -\lambda V & , X \in \bar{\Sigma}_a \\ V(X) = 0 & , X \in S_a := \partial \bar{\Sigma}_a \end{cases} \quad (\star)$$

记 V_{nm} 对应的特征值为 λ_{nm} .

• $\lambda \geq 0$ • $\{V_{nm}\}$ 是 $L^2(\bar{\Sigma}_a)$ 正交的.

即在圆盘上求解该 PDE

将圆盘区域转换为矩形区域: 考虑极坐标 $\bar{V}(r, \theta) = V(r \cos \theta, r \sin \theta)$

$$(\star) \text{ 等价于 } \begin{cases} \frac{1}{r} \partial_r (r \bar{V}_r) + \frac{1}{r^2} \bar{V}_{\theta\theta} = -\lambda \bar{V}, (r, \theta) \in [0, a) \times [0, 2\pi) \\ \bar{V}(a, \theta) = 0, \theta \in [0, 2\pi) \\ \bar{V}(r, \theta) = \bar{V}(r, \theta + 2\pi), (r, \theta) \in [0, a) \times [0, 2\pi). \end{cases} \quad (\bar{\star})$$

对 $(\bar{\star})$ 用分离变量法: $\bar{V}(\theta, r) = H(\theta) R(r)$, 代入 $(\bar{\star})$ 得两组 ODE.

$$(\bar{\star})_a \begin{cases} H''(\theta) + \mu H(\theta) = 0 \\ H(\theta + 2\pi) = H(\theta) \end{cases} \text{ 和 } (\bar{\star})_b \begin{cases} r^2 R''(r) + r R'(r) + (\lambda r^2 - \mu) R(r) = 0 \\ R(a) = 0, |R(r)| < \infty (0 \leq r \leq a) \end{cases}$$

$$(\bar{\star})_a: \mu_n = n^2, H_n(x) = A_n \cos n\theta + B_n \sin n\theta, n = 0, 1, 2, \dots$$

(☆b): 1° $\lambda = 0$, 这是 Euler 方程. $\mu_n = n^2$, 得 $R_{n0}(r) = \begin{cases} a_0 + b_0 \ln r, & n=0 \\ a_n r^n + b_n r^{-n}, & n=1, 2, 3, \dots \end{cases}$

由 $|R_{n0}(r)| < \infty$, $b_n = 0$ ($n \geq 0$). 由 $R_{n0}(a) = 0$, $R_{n0} \equiv 0$.

2° $\lambda > 0$, 考虑变换:

$$x = \sqrt{\lambda} r, \quad y(x) = R\left(\frac{x}{\sqrt{\lambda}}\right)$$

则 (☆b) 转化为:

$$\begin{cases} y''(x) + \frac{y'(x)}{x} + \left(1 - \frac{n^2}{x^2}\right) y(x) = 0, & 0 \leq x \leq a\sqrt{\lambda} \\ y(a\sqrt{\lambda}) = 0, \quad |y| < \infty. \end{cases}$$

这是 n 阶 Bessel 方程 ($n \geq 0$)

用幂级数法解 n 阶 Bessel 方程: $\begin{cases} y'' + \frac{y'}{x} + \left(1 - \frac{n^2}{x^2}\right) y = 0, & (0 \leq x \leq l) \\ y(l) = 0, \quad |y| < \infty \end{cases}$

其中 l 是 Bessel 方程的零点, $\lambda = \frac{l^2}{a^2}$.

其解是如下 n 阶 Bessel 函数:

$$J_n(x) = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!(j+n)!} \left(\frac{x}{2}\right)^{2j+n}.$$

$J_n(x)$ 有零点: $0 < \xi_{n1} < \xi_{n2} < \xi_{n3} < \dots$

$$\therefore \lambda_{nm} = \left(\frac{\xi_{nm}}{a}\right)^2, \quad R_{nm}(r) = J_n\left(\frac{\xi_{nm}}{a} r\right) \quad (n \geq 0, m \geq 1).$$

▷ 给定 $l > 0$, $\{J_n(\alpha_m x)\}_{m \geq 1}$ (其中 $\alpha_m = \frac{\xi_{nm}}{l}$) 带权函数 x 正交 $[L^2([0, l])]$:

$$\int_0^l J_n(\alpha_m x) J_n(\alpha_{m'} x) x dx = 0, \quad \forall m \neq m', \quad m, m' \geq 1$$

▷ n 阶 Bessel 函数的模由 $n+1$ 阶 Bessel 函数的零点值决定:

$$\int_0^l J_n^2(\alpha_m x) x dx = \frac{l^2}{2} [J_n'(\xi_{nm})]^2 = \frac{l^2}{2} J_{n+1}^2(\xi_{nm}) =: \frac{l^2}{2} \beta_{nm}.$$

♠ 综合可知, 问题(★)的特征值与相应的特征函数为

$$\lambda_{nm} = \left(\frac{\xi_{nm}}{a}\right)^2, \quad V_{nm} = (A_{nm} \cos n\theta + B_{nm} \sin n\theta) J_n\left(\frac{\xi_{nm}}{a} r\right), \quad (n \geq 0, m \geq 1).$$

♠ 假设 $u = \sum_{n,m} V_{nm}(x, y) T_{nm}(t)$ 满足(★), 求 T_{nm} 。由 $\{V_{nm}\}$ 的正交性,

$$T'_{nm}(t) + \lambda_{nm} T_{nm}(t) = 0.$$

回忆 $\lambda_{nm} = \left(\frac{\xi_{nm}}{a}\right)^2$ 。因此

$$T_{nm}(t) = e^{-\left(\frac{\xi_{nm}}{a}\right)^2 t}, \quad (n \geq 0, m \geq 1).$$

于是

$$u = \sum_{n=0}^{\infty} \sum_{m=1}^{\infty} (A_{nm} \cos n\theta + B_{nm} \sin n\theta) e^{-\left(\frac{\xi_{nm}}{a}\right)^2 t} J_n\left(\frac{\xi_{nm}}{a} r\right).$$

♠ 下面利用初值条件确定 $\{A_{nm}\}$ 和 $\{B_{nm}\}$:

$$\phi(r, \theta) = \sum_{n=0}^{\infty} \left\{ \left[\sum_{m=1}^{\infty} A_{nm} J_n\left(\frac{\xi_{nm}}{a} r\right) \right] \cos n\theta + \left[\sum_{m=1}^{\infty} B_{nm} J_n\left(\frac{\xi_{nm}}{a} r\right) \right] \sin n\theta \right\}.$$

► 利用 $\{\cos n\theta, \sin n\theta\}_{n \geq 0}$ 的 $L^2([0, 2\pi])$ 正交性,

$$\sum_{m=1}^{\infty} A_{nm} J_n\left(\frac{\xi_{nm}}{a} r\right) = \frac{1}{\pi} \int_0^{2\pi} \phi(r, \theta) \cos n\theta, \quad n = 1, 2, \dots,$$

$$\sum_{m=1}^{\infty} B_{nm} J_n\left(\frac{\xi_{nm}}{a} r\right) = \frac{1}{\pi} \int_0^{2\pi} \phi(r, \theta) \sin n\theta, \quad n = 1, 2, \dots,$$

$$A_{0m} J_0\left(\frac{\xi_{0m}}{a} r\right) = \frac{1}{2\pi} \int_0^{2\pi} \phi(r, \theta), \quad m \geq 1.$$

► 再利用 $\{J_n\left(\frac{\xi_{nm}}{a} r\right)\}$ 在 $[0, a]$ 上带权函数 r 的正交性, 我们得到

$$A_{nm} = \frac{2}{a^2 \pi \beta_{nm}} \int_0^a \left(\int_0^{2\pi} \phi(r, \theta) \cos n\theta d\theta \right) J_n\left(\frac{\xi_{nm}}{a} r\right) r dr, \quad n, m \geq 1,$$

$$B_{nm} = \frac{2}{a^2 \pi \beta_{nm}} \int_0^a \left(\int_0^{2\pi} \phi(r, \theta) \sin n\theta d\theta \right) J_n\left(\frac{\xi_{nm}}{a} r\right) r dr, \quad n, m \geq 1,$$

$$A_{0m} = \frac{1}{a^2 \beta_{0m} \pi} \int_0^r \left(\int_0^{2\pi} \phi(r, \theta) d\theta \right) J_0\left(\frac{\xi_{0m}}{a} r\right) r dr, \quad m \geq 1,$$

其中 $\beta_{nm} = [J'_n(\xi_{nm})]^2 = [J_{n+1}(\xi_{nm})]^2$ 。

求解 n 阶 Bessel 方程: $n = 0, 1, 2, \dots$

$$\begin{cases} y'' + \frac{y'}{x} + (1 - \frac{n^2}{x^2}) y = 0, & (0 \leq x \leq 1) \\ y(1) = 0, & |y| < \infty. \end{cases}$$

Bessel 方程的幂级数解法:

假设 $y(x) = x^\gamma \sum_{k=0}^{\infty} a_k x^k$ ($a_0 \neq 0$), 代入方程:

$$\begin{aligned} 0 &= \sum_{k=0}^{\infty} (k+\gamma)(k+\gamma-1) a_k x^{k+\gamma-2} + \sum_{k=0}^{\infty} (k+\gamma) a_k x^{k+\gamma-2} \\ &\quad - n^2 \sum_{k=0}^{\infty} a_k x^{k+\gamma-2} + \sum_{k=0}^{\infty} a_k x^{k+\gamma} \\ &= (\gamma^2 - n^2) a_0 x^{\gamma-2} + [(1+\gamma)^2 - n^2] a_1 x^{\gamma-1} \\ &\quad + \sum_{k=0}^{\infty} \{ [(k+2+\gamma)^2 - n^2] a_{k+2} + a_k \} x^{k+\gamma} \end{aligned}$$

比较系数, $\gamma = \pm n$, $[(1+\gamma)^2 - n^2] a_1 = 0$

$$[(k+2+\gamma)^2 - n^2] a_{k+2} + a_k = 0.$$

$$|y| < \infty, \quad \gamma = n, \quad a_1 = 0.$$

$$a_{k+2} = - \frac{a_k}{(k+2+\gamma)^2 - n^2} = - \frac{a_k}{(k+2)(k+2n+2)} \quad (\forall k \geq 0)$$

$$\text{易得 } a_{2j+1} = 0 \quad (\forall j \geq 0), \quad a_{2j} = - \frac{a_{2j-2}}{4j(j+n)}$$

$$\frac{a_{2j}}{a_{2j-2}} = - \frac{1}{4j(j+n)}, \quad \frac{a_{2j-2}}{a_{2j-4}} = - \frac{1}{4(j-1)(j+n-1)}, \dots, \quad \frac{a_2}{a_0} = - \frac{1}{4 \cdot 1 \cdot (1+n)}$$

$$\therefore \frac{a_{2j}}{a_0} = \frac{(-1)^j}{4^j \cdot j! \cdot \frac{(j+n)!}{n!}}, \quad a_{2j} = \frac{(-1)^j \cdot n!}{2^{2j} \cdot j! \cdot (j+n)!} \cdot a_0 \quad (\forall j \geq 0).$$

⇒ n 阶 Bessel 方程的解:

$$J_n(x) = \sum_{j=0}^{\infty} \frac{(-1)^j a_0 n!}{2^{2j} \cdot j! \cdot (j+n)!} x^{2j+n}$$

称为 n 阶 Bessel 函数.

为简化, 取 $a_0 = \frac{1}{2^n \cdot n!}$

则

$$J_n(x) = \sum_{j=0}^{\infty} \frac{(-1)^j}{j! \cdot (j+n)!} \left(\frac{x}{2}\right)^{2j+n}$$

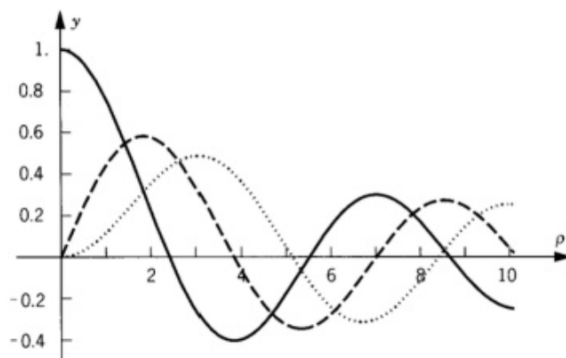
Bessel 函数的零点

n 阶 Bessel 函数是 n 阶 Bessel 方程的解

$$y'' + \frac{y'}{x} + \left(1 - \frac{n^2}{x^2}\right)y = 0.$$

对任何固定的 $n \geq 0$, 当 x 足够大时, 方程的表现接近于 $y'' + y = 0$ 。事实上

$$J_n(x) \simeq \sqrt{\frac{2}{\pi x}} \cos\left(x - \frac{1}{4} - \frac{n}{2}\pi\right) + O(x^{-3/2}).$$



对任何固定的 $n \geq 0$, J_n 有无穷多个正零点:

$$0 < \xi_{n1} < \xi_{n2} < \xi_{n3} < \dots$$

Bessel 函数的正交性

给定 $\ell > 0$, $\{J_n(\alpha_i x)\}_{i \geq 0}$ (其中 $\alpha_i = \frac{\xi_{ni}}{\ell}$) 带权函数 x 正交:

$$\int_0^\ell J_n(\alpha_i x) J_n(\alpha_j x) x dx = 0, \quad \forall i \neq j, \quad i, j \geq 1.$$

证明. 记 $f(x) = J_n(\alpha_i x)$, $g(x) = J_n(\alpha_j x)$. 因为 $J_n(x)$ 是 n 阶 Bessel 方程的解, 故

$$\begin{aligned} x f''(x) + f'(x) + \left(\alpha_i^2 x - \frac{n^2}{x} \right) f &= 0, \\ x g''(x) + g'(x) + \left(\alpha_j^2 x - \frac{n^2}{x} \right) g &= 0. \end{aligned}$$

在第一式两边乘 g , 第二式两边乘 f , 相减后积分得到

$$\begin{aligned} 0 &= \int_0^\ell \left[x f'' g + f' g + \alpha_i^2 x f g - x f g'' - f g' - \alpha_j^2 x f g \right] dx \\ &= (\alpha_i^2 - \alpha_j^2) \int_0^\ell f g x dx + \int_0^\ell \left[x f'' g - x f g'' + f' g - f g' \right] dx \\ &= (\alpha_i^2 - \alpha_j^2) \int_0^\ell f g x dx + [x(f' g - f g')] \Big|_0^\ell \\ &= (\alpha_i^2 - \alpha_j^2) \int_0^\ell f g x dx \quad (\text{利用 } f(\ell) = g(\ell) = 0). \end{aligned}$$

Bessel 函数的模

给定 $\ell > 0$, 取 $\alpha_m = \frac{\xi_{nm}}{\ell}$, 则有

$$\int_0^\ell J_n^2(\alpha_m x) x dx = \frac{\ell^2}{2} [J_n'(\xi_{nm})]^2 = \frac{\ell^2}{2} J_{n+1}^2(\xi_{nm}) =: \frac{\ell^2}{2} \beta_{nm}, \quad \forall m \geq 1.$$

在第二个等式中, 我们使用了恒等式 (可通过 Bessel 函数的级数形式直接验证):

$$J_{n \pm 1}(x) = \frac{n}{x} J_n(x) \mp J_n'(x).$$

证明. 记 $f(x) = J_n(\alpha_m x)$, 我们有 $x f''(x) + f'(x) + x^{-1}(\alpha_m^2 x^2 - n^2) f = 0$, 故

$$0 = 2x f' [(x f')' + x^{-1}(\alpha_m^2 x^2 - n^2) f] = [(x f')^2 + (\alpha_m^2 x^2 - n^2) f^2]' - 2\alpha_m^2 x f^2.$$

对该等式积分, 我们可以得到

$$\begin{aligned} \int_0^\ell J_n^2(\alpha_m x) x dx &= \frac{1}{2\alpha_m^2} \int_0^\ell [(x f')^2 + (\alpha_m^2 x^2 - n^2) f^2]' dx \\ &= \frac{1}{2\alpha_m^2} [(x f')^2 + (\alpha_m^2 x^2 - n^2) f^2] \Big|_0^\ell \\ &= \frac{\ell^2}{2} [J_n'(\xi_{nm})]^2 \quad (\text{使用 } f'(x) = \alpha_m J_n'(\alpha_m x)). \end{aligned}$$

柱域上的电位分布 圆柱体 $C = \{(x, y, z) : x^2 + y^2 \leq a^2, 0 \leq z \leq h\}$ 内没有电荷, 侧面和上底电势为零。下底 $z = 0$ 上电位分布已知。电位函数满足

$$\begin{cases} \Delta u = u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} + u_{zz} = 0, \\ u|_{r=a} = u|_{z=h} = 0, \\ u|_{z=0} = \phi(r, \theta). \end{cases}$$

目标 找解 $u(x, y, z) = \sum_{n,m} V_{nm}(x, y) Z_{nm}(z)$ 。其中 V_{nm} 满足下列特征值问题

$$(\star) \begin{cases} \Delta V = -\lambda V, & (x, y) \in \Sigma_a, \\ V(x, y) = 0, & (x, y) \in S_a = \partial\Sigma_a. \end{cases}$$

记 V_{nm} 对应的特征值为 λ_{nm} 。

♡ 回忆 $\lambda_{nm} \geq 0$, 且 $\{V_{nm}\}$ 是 $L^2(\Sigma_a)$ 正交的。代入方程

$$\begin{aligned} 0 = \Delta u &= \sum_{n,m} Z_{nm} \Delta V_{nm} + \sum_{n,m} V_{nm} Z''_{nm} \\ &= -\sum_{n,m} Z_{nm} \lambda_{nm} V_{nm} + \sum_{n,m} V_{nm} Z''_{nm} \\ \Rightarrow Z''_{nm} &= \lambda_{nm} Z_{nm}. \end{aligned}$$

♠ 回忆: 问题(★)的特征值与相应的特征函数为

$$\lambda_{nm} = \left(\frac{\xi_{nm}}{a}\right)^2, \quad V_{nm} = (A_{nm} \cos n\theta + B_{nm} \sin n\theta) J_n\left(\frac{\xi_{nm}}{a} r\right), \quad (n \geq 0, m \geq 1).$$

♠ 求解 $Z''_{nm} = \lambda_{nm} Z_{nm}$ 得到

$$Z_{nm}(z) = C_{nm} e^{-\frac{\xi_{nm}}{a} z} + D_{nm} e^{\frac{\xi_{nm}}{a} z} = C'_{nm} \cosh z + D'_{nm} \sinh z.$$

利用边值条件 $u(x, y, h) = 0$, 即 $Z_{nm}(h) = 0$, 得到 (相差一个常数乘子)

$$Z_{nm}(z) = \sinh\left(\frac{\xi_{nm}}{a}(z-h)\right), \quad (n \geq 0, m \geq 1).$$

于是

$$u(r, \theta, z) = \sum_{n=0}^{\infty} \sum_{m=1}^{\infty} (A_{nm} \cos n\theta + B_{nm} \sin n\theta) \sinh\left(\frac{\xi_{nm}}{a}(z-h)\right) J_n\left(\frac{\xi_{nm}}{a} r\right).$$

♠ 利用边值条件 $u|_{z=0} = \phi(r, \theta)$ 确定 $\{A_{nm}\}$ 和 $\{B_{nm}\}$: 取 $\gamma_{nm} = -\sinh\left(\frac{\xi_{nm}}{a} h\right)$,

$$\phi(r, \theta) = \sum_{n=0}^{\infty} \left\{ \left[\sum_{m=1}^{\infty} A_{nm} \gamma_{nm} J_n\left(\frac{\xi_{nm}}{a} r\right) \right] \cos n\theta + \left[\sum_{m=1}^{\infty} B_{nm} \gamma_{nm} J_n\left(\frac{\xi_{nm}}{a} r\right) \right] \sin n\theta \right\}.$$

利用 $\{\cos n\theta, \sin n\theta\}_{n \geq 0}$ 的 $L^2([0, 2\pi])$ 正交性, 以及 $\{J_n\left(\frac{\xi_{nm}}{a} x\right)\}_{m \geq 1}$ 带权函数 x 的 $L^2_x([0, a])$ 正交性:

$$\begin{aligned} A_{nm} &= \frac{2}{a^2 \pi \beta_{nm} \gamma_{nm}} \int_0^a \left(\int_0^{2\pi} \phi(r, \theta) \cos n\theta d\theta \right) J_n\left(\frac{\xi_{nm}}{a} r\right) r dr, \quad (n, m \geq 1), \\ B_{nm} &= \frac{2}{a^2 \pi \beta_{nm} \gamma_{nm}} \int_0^a \left(\int_0^{2\pi} \phi(r, \theta) \sin n\theta d\theta \right) J_n\left(\frac{\xi_{nm}}{a} r\right) r dr, \quad (n, m \geq 1), \\ A_{0m} &= \frac{1}{a^2 \beta_{0m} \gamma_{nm} \pi} \int_0^a \left(\int_0^{2\pi} \phi(r, \theta) d\theta \right) J_0\left(\frac{\xi_{0m}}{a} r\right) r dr, \quad m \geq 1, \end{aligned}$$

其中 $\beta_{nm} = [J'_n(\xi_{nm})]^2 = [J_{n+1}(\xi_{nm})]^2$ 。

柱域上的分离变量法: Legendre 多项式的引出

考虑半径为 a 的球 $B_a = \{X = (x, y, z) \in \mathbb{R}^3 : \|X\| < a\}$ 上的 Laplace 方程:

$$\begin{cases} \Delta u(X) = 0, & X \in B_a \\ u(X) = f(X), & X \in S_a := \partial B_a. \end{cases}$$

引入球坐标 $x = r \sin \phi \cos \theta$, $y = r \sin \phi \sin \theta$, $z = r \cos \phi$

$$\bar{u}(r, \theta, \phi) = u(r \sin \phi \cos \theta, r \sin \phi \sin \theta, r \cos \phi)$$

$$\bar{f}(\theta, \phi) = f(a \sin \phi \cos \theta, a \sin \phi \sin \theta, a \cos \phi)$$

球域上的调和方程在球坐标下可以写为:

$$\begin{cases} (r^2 \bar{u}_r)_r + \frac{\bar{u}_{\theta\theta}}{\sin^2 \phi} + \frac{(\sin \phi \bar{u}_\phi)_\phi}{\sin \phi} = 0, & (r, \theta, \phi) \in (0, a) \times [0, 2\pi) \times [0, \pi), \\ \bar{u}(a, \theta, \phi) = \bar{f}(\theta, \phi) & , (\theta, \phi) \in [0, 2\pi) \times [0, \pi), \\ \bar{u}(r, \theta, \phi) = \bar{u}(r, \theta + 2\pi, \phi) & , (r, \theta, \phi) \in (0, a) \times [0, 2\pi) \times [0, \pi). \end{cases}$$

如何在球坐标下计算 Laplace 算子

直接计算得到

$$X_\theta = (x_\theta, y_\theta, z_\theta) = r(-\sin \phi \sin \theta, \sin \phi \cos \theta, 0),$$

$$X_\phi = (x_\phi, y_\phi, z_\phi) = r(\cos \phi \cos \theta, \cos \phi \sin \theta, -\sin \phi),$$

$$X_r = (x_r, y_r, z_r) = (\sin \phi \cos \theta, \sin \phi \sin \theta, \cos \phi).$$

因此

$$(g_{ij}) = \begin{pmatrix} \langle X_\theta, X_\theta \rangle & \langle X_\theta, X_\phi \rangle & \langle X_\theta, X_r \rangle \\ \langle X_\phi, X_\theta \rangle & \langle X_\phi, X_\phi \rangle & \langle X_\phi, X_r \rangle \\ \langle X_r, X_\theta \rangle & \langle X_r, X_\phi \rangle & \langle X_r, X_r \rangle \end{pmatrix} = \begin{pmatrix} r^2 \sin^2 \phi & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

利用 Laplace 算子在一般坐标下的表达: (取 $\xi_1 = \theta$, $\xi_2 = \phi$, $\xi_3 = r$)

$$\begin{aligned} \Delta u &= \frac{1}{\sqrt{\det g}} \frac{\partial}{\partial \xi_i} \left(\sqrt{\det g} g^{ij} \frac{\partial}{\partial \xi_j} \bar{u} \right) \\ &= \frac{1}{r^2 \sin \phi} \left[\frac{\partial}{\partial \theta} \left(\frac{1}{\sin \phi} \frac{\partial}{\partial \theta} \bar{u} \right) + \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial}{\partial \phi} \bar{u} \right) + \frac{\partial}{\partial r} \left(r^2 \sin \phi \frac{\partial}{\partial r} \bar{u} \right) \right] \\ &= \frac{\bar{u}_{\theta\theta}}{r^2 \sin^2 \phi} + \frac{1}{r^2 \sin \phi} \frac{\partial}{\partial \phi} \left(\sin \phi \frac{\partial}{\partial \phi} \bar{u} \right) + \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \bar{u} \right). \end{aligned}$$

分离变量法 找解 $\bar{u}(r, \theta, \phi) = \sum_{n,m} H_n(\theta) V_{nm}(r, \phi)$ 。其中 H_n 满足特征值问题

$$\begin{cases} H'' + \mu H = 0, \\ H(\theta) = H(\theta + 2\pi). \end{cases}$$

回忆该特征值问题的解为

$$\mu_n = n^2, \quad H_n = A_n \cos n\theta + B_n \sin n\theta, \quad n = 0, 1, 2, \dots$$

代入方程得到

$$(r^2 \partial_r V_{nm})_r H_n - \frac{\mu_n H_n V_{nm}}{\sin^2 \phi} + \frac{(\sin \phi \partial_\phi V_{nm})_\phi}{\sin \phi} H_n = 0.$$

利用 $\{H_n\}_{n \geq 0}$ 的 L^2 正交性, V_{nm} 满足方程:

$$(\star) \quad (r^2 V_r)_r + \frac{(\sin \phi V_\phi)_\phi}{\sin \phi} = \frac{n^2 V}{\sin^2 \phi}.$$

我们仍然利用分离变量求上述方程。为此, 假设 $V(r, \phi) = R(r)G(\phi)$ 。代入 $(\star)$ 可知

$$\frac{(r^2 R_r)_r}{R} = \frac{n^2}{\sin^2 \phi} - \frac{(\sin \phi G_\phi)_\phi}{G \sin \phi} = \lambda.$$

因此 $(\star)$ 被分离为两个常微分方程

$$(\star)_1 \quad \frac{(\sin \phi G')'}{\sin \phi} + \left(\lambda - \frac{n^2}{\sin^2 \phi} \right) G = 0 \quad \text{和} \quad (\star)_2 \quad (r^2 R')' = \lambda R.$$

为研究 $(\star)_1$, 引入变量 $s = \cos \phi$, 并令 $\bar{G}(s) = G(\arccos s)$ 。利用 $ds = -\sin \phi d\phi$ 得

$$(\bar{\star})_1 \quad \begin{cases} \frac{d}{ds} \left[(1-s^2) \frac{d\bar{G}}{ds} \right] + \left(\lambda - \frac{n^2}{1-s^2} \right) \bar{G}(s) = 0, & s \in [-1, 1], \\ |\bar{G}(s)| < \infty. \end{cases}$$

- **Legendre 方程:** $(1-x^2)y'' - 2xy' + \lambda y = 0, \quad x \in [-1, 1].$
 - m 阶 **Legendre 方程:** $(1-x^2)y'' - 2xy' + m(m+1)y = 0, \quad x \in [-1, 1].$
 - 伴随 **Legendre 方程:** $(1-x^2)y'' - 2xy' + \left(\lambda - \frac{n^2}{1-x^2} \right) y = 0, \quad x \in [-1, 1].$
- 容易验证, $(\bar{\star})_1$ 是伴随 Legendre 方程。

问题: 如何求解伴随 Legendre 方程?

我们将用幂级数法求解伴随 Legendre 方程

$$\begin{cases} (1-x^2)y'' - 2xy' + \left(\lambda - \frac{n^2}{1-x^2}\right)y = 0, & x \in [-1, 1], \\ |y(x)| < \infty. \end{cases}$$

其特征值是 $\lambda = m(m+1)$ ($m \geq n$)，解是 m 次 n 阶伴随 Legendre 多项式：

$$P_m^n(x) = (1-x^2)^{\frac{n}{2}} \frac{d^n}{dx^n} P_m(x), \quad (m \geq n \geq 0),$$

其中 $P_m(x)$ 是如下 m 次 Legendre 多项式，它是 m 阶 Legendre 方程的解，

$$P_m(x) = \frac{1}{2^m m!} \frac{d^m}{dx^m} (x^2-1)^m, \quad (m = 0, 1, 2, \dots).$$

► 伴随 Legendre 多项式的 $L^2([-1, 1])$ 正交性：

$$\int_{-1}^1 P_m^n(x) P_{\bar{m}}^{\bar{n}}(x) dx = 0, \quad \forall m \neq \bar{m}, \quad m, \bar{m}, n \geq 0.$$

► 伴随 Legendre 多项式的模：

$$\int_{-1}^1 [P_m^n(x)]^2 dx = \frac{2}{2m+1} \frac{(m+n)!}{(m-n)!} =: p_m^n, \quad \forall m \geq n \geq 0.$$

♠ 通过求解 (伴随)Legendre 方程，问题(★)₁的特征值和特征函数为

$$\lambda_m = m(m+1), \quad G_{nm}(\phi) = P_n^m(\cos \phi), \quad (n \geq 0, m \geq n).$$

♠ 求解(★)₂ $(r^2 R'_m)' = \lambda_m R_m$ 。容易看到 $R_m = C_1 r^{\alpha_m} + C_2 r^{-\beta_m}$ 。代入知

$$R_m = C_1 r^m + C_2 r^{-m-1}.$$

为使 $|R_m(r)| < \infty$ ，我们取

$$R_m(r) = r^m.$$

♠ 综上可知问题(★)的解为

$$V_{nm}(r, \phi) = C_{nm} P_n^m(\cos \phi) r^m, \quad (n \geq 0, m \geq n).$$

这样，便得到了分离变量解的形式

$$\bar{u}(r, \theta, \phi) = \sum_{m=0}^{\infty} \sum_{n=0}^m r^m (A_{nm} \cos n\theta + B_{nm} \sin n\theta) P_n^m(\cos \phi).$$

♠ 通过边值条件确定 A_{nm} 和 B_{nm} :

$$\bar{f}(\theta, \phi) = \sum_{m=0}^{\infty} \sum_{n=0}^m a^m (A_{nm} \cos n\theta + B_{nm} \sin n\theta) P_m^n(\cos \phi).$$

利用 $\{\cos n\theta, \sin n\theta\}_{n \geq 0}$ 的 $L^2([0, 2\pi])$ 正交性:

$$\int_0^{2\pi} \bar{f}(\theta, \phi) \cos n\theta d\theta = \pi \sum_{m \geq n} a^m A_{nm} P_m^n(\cos \phi),$$

$$\int_0^{2\pi} \bar{f}(\theta, \phi) \sin n\theta d\theta = \pi \sum_{m \geq n} a^m B_{nm} P_m^n(\cos \phi).$$

再用 $\{G_{nm}(\phi) = P_m^n(\cos \phi)\}_{m \geq n \geq 0}$ 带权函数 $\sin \phi$ 的 $L^2_{\sin \phi}([0, \pi])$ 正交性:

$$\int_0^{\pi} \left[\int_0^{2\pi} \bar{f}(\theta, \phi) \cos n\theta d\theta \right] P_m^n(\cos \phi) \sin \phi d\phi = \pi a^m p_m^n A_{nm},$$

$$\int_0^{\pi} \left[\int_0^{2\pi} \bar{f}(\theta, \phi) \sin n\theta d\theta \right] P_m^n(\cos \phi) \sin \phi d\phi = \pi a^m p_m^n B_{nm}.$$

求解 m 阶 Legendre 方程 ($m=0, 1, 2, \dots$)

取 $\lambda = m(m+1)$.

$$\begin{cases} (1-x^2)y'' - 2xy' + \lambda y = 0, & x \in [-1, 1] \\ |y(x)| < \infty \end{cases}$$

假设有幂级数解: $y(x) = \sum_{k=0}^{\infty} a_k x^k$.

$$= \sum_{k=2}^{\infty} a_k \cdot k \cdot (k-1) x^{k-2} = \sum_{k'=0}^{\infty} a_{k'+2} \cdot (k'+2)(k'+1) \cdot x^{k'}$$

$$0 = (1-x^2) \sum_{k=0}^{\infty} a_k (k-1)k x^{k-2} - 2x \sum_{k=0}^{\infty} a_k k x^{k-1} + \lambda \sum_{k=0}^{\infty} a_k x^k$$

$$= \sum_{k=0}^{\infty} (k-1)k a_k x^{k-2} - \sum_{k=0}^{\infty} (k^2 + k - \lambda) a_k x^k$$

$$= \sum_{k=0}^{\infty} a_{k+2} \cdot (k+1)(k+2) x^k - \sum_{k=0}^{\infty} (k^2 + k - \lambda) a_k x^k$$

$$\text{比较系数: } a_{k+2} = \frac{k(k+1) - \lambda}{(k+1)(k+2)} a_k = - \frac{(m-k)(k+m+1)}{(k+1)(k+2)} a_k, \quad k=0, 1, 2, \dots$$

易得 $a_{m+2} = a_{m+4} = a_{m+6} = \dots = 0$.

更一般地, a_{2k} 由 a_0 确定:

$$a_2 = - \frac{m(m+1)}{2!} a_0$$

$$a_4 = \frac{(m-2)m(m+1)(m+3)}{4!} a_0$$

$$a_6 = - \frac{(m-4)(m-2)m(m+1)(m+3)(m+5)}{6!} a_0$$

$$a_{2k} = (-1)^k \cdot \frac{\prod_{l=1}^k (m-1+2l) \cdot \prod_{l=1}^k (m+2-2l)}{(2k)!} a_0 \quad (k \geq 1)$$

同样地, a_{2k+1} 由 a_1 确定:

$$a_{2k+1} = (-1)^k \cdot \frac{\prod_{l=1}^k (m+2l) \cdot \prod_{l=1}^k (m+1-2l)}{(2k+1)!} a_1 \quad (k \geq 1)$$

则幂级数可写为 $y(x) = a_0 y_{m_0}(x) + a_1 y_{m_1}(x)$.

$$\text{其中 } y_0(x) = 1 + \sum_{k \geq 1} (-1)^k \cdot \frac{\prod_{l=1}^k (m-1+2l) \cdot \prod_{l=1}^k (m+2-2l)}{(2k)!} x^{2k}$$

$$y_1(x) = x + \sum_{k \geq 1} (-1)^k \cdot \frac{\prod_{l=1}^k (m+2l) \cdot \prod_{l=1}^k (m+1-2l)}{(2k+1)!} x^{2k+1}$$

若 m 是奇数, $y_{m_1}(x)$ 是 m 阶多项式. 记 $P_m(x) = \frac{y_{m_1}(x)}{y_{m_1}(1)}$.

$$P_m(x) = \frac{1}{y_{m_1}(1)} \left[x + \sum_{k=1}^{\frac{m-1}{2}} (-1)^k \cdot \frac{\prod_{l=1}^k (m+2l) \cdot \prod_{l=1}^k (m+1-2l)}{(2k+1)!} x^{2k+1} \right]$$

称为 m 阶 Legendre 多项式 (即取 $a_1 = \frac{1}{y_{m_1}(1)}$, $a_0 = 0$)

此时 $|y_{m_0}(\pm 1)| = \infty$.

若 m 是偶数, $y_{m_0}(x)$ 是 m 阶多项式. 记 $P_m(x) = \frac{y_{m_0}(x)}{y_{m_0}(1)}$.

$$P_m(x) = \frac{1}{y_{m_0}(1)} \left[x + \sum_{k=1}^{\frac{m}{2}} (-1)^k \cdot \frac{\prod_{l=1}^k (m-1+2l) \cdot \prod_{l=1}^k (m+2-2l)}{(2k)!} x^{2k} \right]$$

称为 m 阶 Legendre 多项式 (即取 $a_0 = \frac{1}{y_{m_0}(1)}$, $a_1 = 0$)

此时 $|y_{m_0}(\pm 1)| = \infty$.

考虑一般的 λ , 也可以写下 Legendre 方程的幂级数解 $y(x)$

但如果 $\lambda \neq m(m+1)$ ($m \geq 0$), 则 y_λ 在 $x = \pm 1$ 处发散.

因此 Legendre 方程 $\begin{cases} (1-x^2)y'' - 2xy' + \lambda y = 0, & x \in [-1, 1] \\ |y(x)| < \infty, & x \in [-1, 1] \end{cases}$

的特征值是 $\lambda_m = m(m+1)$,

特征函数是 m 次 Legendre 多项式 $P_m(x)$ ($m = 0, 1, 2, \dots$)

Rodrigues 公式: Legendre 多项式的微分表达式

$$P_m(x) = \frac{1}{2^m m!} \frac{d^m}{dx^m} (x^2 - 1)^m, \quad (m = 0, 1, 2, \dots).$$

证明. 记 $f_m(x) = (x^2 - 1)^m$. 则 $f'_m = 2mx(x^2 - 1)^{m-1}$, 于是

$$(x^2 - 1)f'_m = 2mxf_m.$$

两边对 x 求 $m+1$ 阶导数, 并利用 Leibniz 公式: $(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(n-k)} v^{(k)}$,

$$(x^2 - 1)f_m^{(m+2)} + 2(m+1)xf_m^{(m+1)} + m(m+1)f_m^{(m)} = 2mxf_m^{(m+1)} + 2m(m+1)f_m^{(m)}.$$

化简表明

$$(x^2 - 1)f_m^{(m+2)} + 2xf_m^{(m+1)} = m(m+1)f_m^{(m)}.$$

即 $f_m^{(m)}$ 是 Legendre 方程的解. 另一个方面, 幂级数解法告诉我们:

Legendre 方程的有界解 (即 Legendre 多项式) 与 $f_m^{(m)}$ 只相差一个 (常数) 乘数.

容易验证 $f_m^{(m)}(1) = 2^m m!$, 所以 $P_m(x) = \frac{1}{2^m m!} f_m^{(m)}(x)$, 证毕.

Legendre 多项式的递推关系

$$(m+1)P_{m+1}(x) - (2m+1)xP_m(x) + mP_{m-1}(x) = 0.$$

证明. 利用 Rodrigues 公式,

$$\begin{aligned}P_{m+1}(x) &= \frac{1}{2^{m+1}(m+1)!} \frac{d^{m+1}}{dx^{m+1}} (x^2 - 1)^{m+1} \\&= \frac{1}{2^m m!} \frac{d^m}{dx^m} \left[x(x^2 - 1)^m \right] \\&= \frac{1}{2^m m!} \left[x \frac{d^m}{dx^m} (x^2 - 1)^m + m \frac{d^{m-1}}{dx^{m-1}} (x^2 - 1)^m \right] \\&= xP_m(x) + \frac{m}{2^m m!} \frac{d^{m-1}}{dx^{m-1}} (x^2 - 1)^m \quad (1).\end{aligned}$$

两边关于 x 求一次导数:

$$P'_{m+1}(x) = xP'_m(x) + (m+1)P_m(x) \quad (2).$$

另一方面, 还是通过 Rodrigues 公式,

$$\begin{aligned}P_{m+1}(x) &= \frac{1}{2^{m+1}(m+1)!} \frac{d^{m+1}}{dx^{m+1}} [(x^2 - 1)^m (x^2 - 1)] \\&= \frac{1}{2^{m+1}(m+1)!} \left[(x^2 - 1) \frac{d^{m+1}}{dx^{m+1}} (x^2 - 1)^m \right. \\&\quad \left. + 2(m+1)x \frac{d^m}{dx^m} (x^2 - 1)^m + m(m+1) \frac{d^{m-1}}{dx^{m-1}} (x^2 - 1)^m \right] \\&= \frac{(x^2 - 1)}{2(m+1)} P'_m(x) + xP_m(x) + \frac{m}{2^{m+1}m!} \frac{d^{m-1}}{dx^{m-1}} (x^2 - 1)^m \quad (3).\end{aligned}$$

(3) $\times 2 - (1)$ 得

$$P_{m+1} = \frac{x^2 - 1}{m+1} P'_m + xP_m \implies P'_m = \frac{m+1}{x^2 - 1} P_{m+1} - \frac{m+1}{x^2 - 1} xP_m \quad (4).$$

把 (2) 中的 $m+1$ 换成 m , 并利用 (4) 计算导数项, 我们得到

$$\begin{aligned}mP_{m-1} &= P'_m - xP'_{m-1} \\&= \frac{m+1}{x^2 - 1} P_{m+1} - \frac{(2m+1)x}{x^2 - 1} P_m + \frac{mx^2}{x^2 - 1} P_{m-1}.\end{aligned}$$

整理即得预期的递推关系式。

Legendre 多项式正交性

$$\int_{-1}^1 P_m(x) P_{\bar{m}}(x) dx = 0, \quad \forall m \neq \bar{m}, \quad m, \bar{m} \geq 0.$$

证明. Legendre 多项式 P_m 和 $P_{\bar{m}}$ 满足 Legendre 方程

$$[(1-x^2)P'_m]' + m(m+1)P_m = 0,$$

$$[(1-x^2)P'_{\bar{m}}]' + \bar{m}(\bar{m}+1)P_{\bar{m}} = 0.$$

第一式 $\times P_{\bar{m}}$ 与第二式 $\times P_m$ 相减后积分:

$$\begin{aligned} (m - \bar{m})(m + \bar{m} + 1) \int_{-1}^1 P_m P_{\bar{m}} &= \int_{-1}^1 \left[[(1-x^2)P'_{\bar{m}}]' P_m - [(1-x^2)P'_m]' P_{\bar{m}} \right] \\ &= \left[(1-x^2)P'_{\bar{m}} P_m \right] \Big|_{-1}^1 - \left[(1-x^2)P'_m P_{\bar{m}} \right] \Big|_{-1}^1 \\ &= 0. \end{aligned}$$

Legendre 多项式的模

$$\int_{-1}^1 P_m^2(x) dx = \frac{2}{2m+1} =: p_m, \quad \forall m \geq 0.$$

证明. 回忆递推公式

$$mP_m(x) - (2m-1)xP_{m-1}(x) + (m-1)P_{m-2}(x) = 0,$$

$$(m+1)P_{m+1}(x) - (2m+1)xP_m(x) + mP_{m-1}(x) = 0.$$

用 $\langle \cdot, \cdot \rangle_{L^2}$ 表示 $[-1, 1]$ 上函数的 L^2 内积. 两式分别与 P_m 、 P_{m-1} 做内积得

$$m\langle P_m, P_m \rangle_{L^2} = (2m-1)\langle P_m, xP_{m-1} \rangle_{L^2}$$

$$m\langle P_{m-1}, P_{m-1} \rangle_{L^2} = (2m+1)\langle xP_m, P_{m-1} \rangle_{L^2}.$$

我们于是得到递推公式

$$(2m+1)\langle P_m, P_m \rangle_{L^2} = (2m-1)\langle P_{m-1}, P_{m-1} \rangle_{L^2} = \cdots = \langle P_0, P_0 \rangle_{L^2} = 2.$$

求解伴随 Legendre 方程

给定 $m \geq n \geq 0$, 考虑伴随 Legendre 方程

$$\begin{cases} (1-x^2)y'' - 2xy' + \left(m(m+1) - \frac{n^2}{1-x^2}\right)y = 0, & x \in [-1, 1], \\ |y(x)| < \infty. \end{cases}$$

可验证, 如下 n 阶 m 次伴随 Legendre 多项式是上述方程的解,

$$P_m^n(x) = (1-x^2)^{\frac{n}{2}} \frac{d^n}{dx^n} P_m(x), \quad (m \geq n \geq 0).$$

证明. 对 $(1-x^2)P_m'' - 2xP_m' + m(m+1)P_m = 0$ 求 n 阶导, 并利用 Leibniz 公式

$$0 = (1-x^2)[P_m^{(n)}]'' - 2(n+1)x[P_m^{(n)}]' + [m(m+1) - n(n+1)]P_m^{(n)}.$$

把 $P_m^{(n)} = (1-x^2)^{-\frac{n}{2}} P_m^n$ 代入上式, 可得

$$0 = (1-x^2)^{-\frac{n}{2}} \left[(1-x^2)[P_m^n]'' - 2x[P_m^n]' + \left(m(m+1) - \frac{n^2}{1-x^2}\right)P_m^n \right].$$

伴随 Legendre 多项式的正交性

$$\int_{-1}^1 P_m^n(x) P_{\bar{m}}^n(x) dx = 0, \quad \forall m \neq \bar{m}, \quad m, \bar{m}, n \geq 0.$$

证明. P_m^n 和 $P_{\bar{m}}^n$ 满足伴随 Legendre 方程

$$[(1-x^2)[P_m^n]']' + \left(m(m+1) - \frac{n^2}{1-x^2}\right)P_m^n = 0,$$

$$[(1-x^2)[P_{\bar{m}}^n]']' + \left(\bar{m}(\bar{m}+1) - \frac{n^2}{1-x^2}\right)P_{\bar{m}}^n = 0.$$

第一式 $\times P_{\bar{m}}^n$ 与第二式 $\times P_m^n$ 相减后积分:

$$\begin{aligned} & (m - \bar{m})(m + \bar{m} + 1) \int_{-1}^1 P_m^n P_{\bar{m}}^n \\ &= \int_{-1}^1 \left[[(1-x^2)[P_{\bar{m}}^n]']' P_m^n - [(1-x^2)[P_m^n]']' P_{\bar{m}}^n \right] \\ &= \left[(1-x^2)[P_{\bar{m}}^n]' P_m^n \right]_{-1}^1 - \left[(1-x^2)[P_m^n]' P_{\bar{m}}^n \right]_{-1}^1 \\ &= 0. \end{aligned}$$

伴随 Legendre 多项式的模

$$\int_{-1}^1 [P_m^n(x)]^2 dx = \frac{2}{2m+1} \frac{(m+n)!}{(m-n)!} =: p_m^n, \quad \forall m \geq n \geq 0.$$

证明. 对递推公式 $(m+1)P_{m+1}(x) - (2m+1)xP_m(x) + mP_{m-1}(x) = 0$ 求 n 阶导

$$\begin{aligned} 0 &= (1-x^2)^{\frac{n}{2}} \left[(m+1)P_{m+1}^{(n)} - (2m+1)xP_m^{(n)} - (2m+1)nP_m^{(n-1)} + mP_{m-1}^{(n)} \right] \\ &= (m+1)P_{m+1}^n - (2m+1)xP_m^n - (2m+1)n\sqrt{1-x^2}P_m^{(n-1)} + mP_{m-1}^n. \end{aligned}$$

另一方面, 把 $\frac{1}{2^{m+1}(m+1)!} \frac{d^m}{dx^m}$ 作用到下面的恒等式

$$\begin{aligned} \frac{d^2}{dx^2}(x^2-1)^{m+1} &= 2(m+1)(x^2-1)^m + 4m(m+1)x^2(x^2-1)^{m-1} \\ &= 2(2m+1)(m+1)(x^2-1)^m + 4m(m+1)(x^2-1)^{m-1}. \end{aligned}$$

我们可以得到 $P'_{m+1} = (2m+1)P_m + P'_{m-1}$. 作用 $(1-x^2)^{\frac{n}{2}} \frac{d^{n-1}}{dx^{n-1}}$ 得

$$P_{m+1}^n = (2m+1)\sqrt{1-x^2}P_m^{(n-1)} + P_{m-1}^n.$$

消去红字项, 我们得到伴随 Legendre 多项式的递推公式

$$0 = (m-n+1)P_{m+1}^n - (2m+1)xP_m^n + (m+n)P_{m-1}^n.$$

分别与 P_{m+1}^n 和 P_{m-1}^n 做 L^2 -内积得到

$$0 = (m-n+1) \int_{-1}^1 [P_{m+1}^n]^2 - (2m+1) \langle xP_m^n, P_{m+1}^n \rangle_{L^2},$$

$$0 = -(2m+1) \langle xP_m^n, P_{m-1}^n \rangle_{L^2} + (m+n) \int_{-1}^1 [P_{m-1}^n]^2.$$

第一式的 m 用 $m-1$ 替换, 再利用第二式消去 $\langle xP_m^n, P_{m-1}^n \rangle_{L^2}$,

$$(2m+1) \int_{-1}^1 [P_m^n]^2 = \frac{m+n}{m-n} (2m-1) \int_{-1}^1 [P_{m-1}^n]^2.$$

利用该递推公式可求得 P_m^n 的 L^2 模。

考试要求: ODE 的幂级数解法

eg. $y'' = xy$.

假设 $y(x) = \sum_{n=0}^{\infty} C_n x^n$ 则 $y'(x) = \sum_{n=1}^{\infty} n C_n x^{n-1}$,

$$y''(x) = \sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} = \sum_{n=0}^{\infty} (n+2)(n+1) C_{n+2} x^n$$

$$\therefore \sum_{n=0}^{\infty} (n+2)(n+1) C_{n+2} x^n = \sum_{n=0}^{\infty} C_n x^{n+1} = \sum_{n=0}^{\infty} C_{n-1} x^n, \text{ 其中 } C_{-1} = 0.$$

$$\therefore (n+2)(n+1) C_{n+2} = C_{n-1} \quad (\forall n \geq 0)$$

隔3递推: $n=0, 3, 6, \dots : C_{-1} = C_2 = C_5 = \dots = 0;$

$$n=1, 4, 7, \dots : C_3 = \frac{1}{3!} C_0, C_6 = \frac{4}{6!} C_0, C_9 = \frac{4 \times 7}{9!} C_0 \dots$$

$$n=2, 5, 8, \dots : C_4 = \frac{2}{4!} C_1, C_7 = \frac{2 \times 5}{7!} C_1, C_{10} = \frac{2 \times 5 \times 8}{10!} C_1 \dots$$

若 C_0, C_1 给定, 则由通项公式可确定所有 $C_n (n \geq 2)$

归纳法: $C_{3k-1} = 0 \quad (k \geq 0)$

$$C_{3k} = \frac{\prod_{l=1}^k (3l-2)}{(3k)!} C_0, C_{3k+1} = \frac{\prod_{l=1}^k (3l-1)}{(3k+1)!} C_1 \quad (k \geq 1)$$

方程的幂级数解为:

$$y(x) = C_0 \left[1 + \sum_{k=1}^{\infty} \frac{\prod_{l=1}^k (3l-2)}{(3k)!} x^{3k} \right] + C_1 \left[x + \sum_{k=1}^{\infty} \frac{\prod_{l=1}^k (3l-1)}{(3k+1)!} x^{3k+1} \right]$$

ex.
$$\begin{cases} y'' - 2xy' + 10y = 0 \\ y(0) = 0, y'(0) = 1 \end{cases}$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^n, \quad y'(x) = \sum_{n=0}^{\infty} n a_n x^{n-1}, \quad y''(x) = \sum_{n=0}^{\infty} (n-1)n a_n x^{n-2}.$$

$$\therefore \sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} x^n - 2 \sum_{n=0}^{\infty} n a_n x^n + 10 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\therefore (n+1)(n+2) a_{n+2} - 2n a_n + 10 a_n = 0$$

$$a_{n+2} = \frac{2n-10}{(n+1)(n+2)} a_n, \quad (\forall n \geq 0)$$

$$n=5, a_7=0, \text{ then } a_7=a_9=a_{11}=\dots=0.$$

$$y(0)=a_0=0, \quad y'(0)=a_1=1$$

$$\therefore a_{2k}=0 \quad (\forall k \geq 0), \quad a_3 = -\frac{4}{3}, \quad a_5 = \frac{15}{4}$$

$$\therefore y = x - \frac{4}{3}x^3 + \frac{15}{4}x^5.$$

ex. $x=0$ 的邻域内用幂级数法求解:

$$(1-x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0.$$

$$y(x) = \sum_{n=0}^{\infty} a_n x^n, \quad (1-x^2) \cdot \sum_{n=2}^{\infty} (n-1)n a_n x^{n-2} + x \sum_{n=1}^{\infty} n a_n x^{n-1} - \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\therefore \sum_{n=0}^{\infty} (n+1)(n+2) a_{n+2} x^n - \sum_{n=0}^{\infty} (n-1)n a_n x^n + \sum_{n=0}^{\infty} n a_n x^n - \sum_{n=0}^{\infty} a_n x^n = 0.$$

$$(n+1)(n+2) a_{n+2} - (n-1)^2 a_n = 0, \quad a_{n+2} = \frac{(n-1)^2}{(n+1)(n+2)} a_n$$

$$n=1, a_3=0, \quad a_{2k+1}=0 \quad (\forall k \geq 1).$$

$$a_2 = \frac{(-1)^2}{2!} a_0, \quad a_4 = \frac{1^2 \cdot (-1)^2}{4!} a_0, \quad a_6 = \frac{3^2 \cdot 1^2 \cdot (-1)^2}{6!} a_0,$$

$$a_{2k} = \frac{[(2k-3)!!]^2}{(2k)!} a_0 \quad (k \geq 1)$$

$$\therefore y = a_1 x + a_0 \left[1 + \sum_{k=1}^{\infty} \frac{[(2k-3)!!]^2}{(2k)!} x^{2k} \right].$$